S5 Glossary

Terms specific to this article, plus carried-over NDT/RVC/EAP terms used throughout

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S5.0 Terms new to this article

Term	Definition
Onset-phase vs. peak-phase	Two distinct reference points within a detected spindle event at which cardiac/respiratory phase can be measured: `onset_sec` (event start, used by the companion RVC_SLEEP article's Mal B) vs. `peak_sec` (the time of maximum amplitude within the event, the reference point this article shows carries the population-level coupling).
Event-timing pathway	The hypothesis that body phase helps determine **when** an endogenous RVC event (or a specific feature of it, e.g. its amplitude peak) occurs - operationalized here as non-uniformity of event phase relative to the cardiac/respiratory cycle (R2-R5).
Event-efficacy pathway	The logically separate hypothesis that, given an event has occurred, the body's phase at that moment changes **how strongly** the event reconfigures state-space geometry - operationalized here as the alignment-moderation test (R6).
Subject-rotation permutation null	A Monte Carlo null distribution constructed by adding one i.i.d. $\mathrm{Uniform}(0, 2\pi)$ rotation per subject to all of that subject's phases, repeated n_{perm} times; destroys any common population-level preferred phase while exactly preserving each subject's event count and within-subject clustering. Replaces an earlier, retracted subject-label-shuffle proposal that left the pooled statistic invariant.
Harmonic statistic (R_1, R_2, T_2)	R_1: first-harmonic mean resultant length (classical Rayleigh concentration). R_2: second-harmonic mean resultant length (sensitive to axial/bimodal structure). $T_2 = R_1^2 + R_2^2$: an omnibus statistic combining both, used throughout this article's permutation tests.
Duration-tercile stratification	Splitting events into three equal-sized groups by duration (short/mid/long) and re-running a test independently within each group, to check whether an effect is duration-general or concentrated in a duration subrange (e.g. a tail artifact).
Leave-one-subject-out (LOSO) jackknife	Re-running a population-level test 24 times, once per dropped subject, to check that no single subject's data disproportionately drives the pooled result.
Deterministic split-half replication	Splitting subjects into two fixed, non-overlapping halves (here: alphabetically first 12 vs. last 12) and re-running a test independently in each half - the strongest single check here for whether an effect is population-general.
Held-out estimation	Estimating a population parameter (here, $\mu_{\mathrm{resp,peak}}$) on a data subset that excludes the events subsequently used to test a hypothesis involving that parameter, to avoid circularity/data leakage.
(Peak-phase) alignment	$\mathrm{alignment}_e = \cos(\varphi_e - \mu)$ for a given event e's phase φ_e and a reference (population-preferred) phase μ: $+1$ when e's phase exactly matches μ, -1 when diametrically opposite, used as a continuous moderator in R6′s regression.
Alignment moderation	Testing whether an event's alignment (above) to a population-preferred phase predicts the size of a separate, downstream outcome for that same event (here, the spindle-locked geometry delta) - distinct from testing whether phase itself is non-uniformly distributed (a timing test).
Geometry-optimal phase	In the secondary circular-regression check (S1.5, S2.5), the phase direction $\mathrm{atan2}(\mathrm{sin_coef}, \mathrm{cos_coef})$ implied by a free two-coefficient fit as maximizing the fitted geometry delta - compared against the independently-estimated timing-preferred phase $\mu_{\mathrm{resp,peak}}$, not assumed to coincide with it a priori.

Figure 1: Terms introduced or newly operationalized in this article.

S5.1 Carried-over NDT/RVC/EAP terms

Term	Definition
Noetic Diffusion Theory (NDT)	The parent geometric framework for neural state dynamics within which RVC and EAP are formulated ((Langell, 2025)); not itself tested by this article.
Rhythmic Variance Control (RVC)	The Tier-1 NDT mechanism under which endogenous, rhythm-defined events (e.g. sleep spindles) are treated as candidate transient gating-precision-change windows.
Embodied Anchoring Principle (EAP)	The Tier-2 NDT extension proposing that peripheral bodily rhythms (cardiac, respiratory) can additionally condition the same latent landscape RVC events act on; this article's R6 is a test relevant to (but not dispositive of) this principle in its event-efficacy form.
`d`, `d_n`, `d_l`, `d_s`, `e`, `e_e`, `e_s`, `e_m`, `m`	The 9 primary geometry endpoints inherited from the companion RVC_SLEEP article's `coords_9d` state-space representation; not redefined here (see companion article's `S1`/`S5` for full definitions).
`event_control` vs. `event_prefar` delta mode	Two ways of computing a geometry "delta" for a given event: `event_control` = event value minus a matched control-window median; `event_prefar` = event value minus the same event's own pre-spindle baseline window ("far") value - both inherited unchanged from the companion article's Mal C machinery.
WLS-cluster (fit_wls_cluster)	Weighted least squares with weights $1/n_{\text{events,subject}}$ (equalizing each subject's total contribution regardless of event count) and cluster-robust standard errors by subject; the shared regression machinery for all moderation tests in both this article and its companion.
BH-FDR	Benjamini-Hochberg false discovery rate correction ((Benjamini & Hochberg, 1995)), applied within each pre-specified test family (4-test primary onset family, S1.3; 18-test alignment-moderation family, S1.5) but not across families, matching the companion article's own convention.

Figure 2: Terms inherited from the companion RVC_SLEEP article and the broader NDT framework, retained here without modification unless otherwise noted.

Bibliography

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